

Q.1 (D)

$f(x, y) = 0$ represents two straight lines which passing through (2, 1), (1, 3), (3, 6) and (0, 0) and no three of these points are collinear.

$$\left. \frac{dy}{dx} \right|_{(2,1)} = \left. \frac{dy}{dx} \right|_{(3,6)}$$

$\therefore$ one of the line is passing (2, 1) and (3, 6) and other line is passing through (0, 0) and (1, 3).

$$\therefore \left. \frac{dy}{dx} \right|_{(0,0)} = 3.$$

Q.2 (D)

$$\lim_{x \rightarrow \infty} \frac{2 - \frac{\ln^2 x}{x}}{\frac{\ln x}{\sqrt{x}} + 1}$$

$$\text{Now, } \lim_{x \rightarrow \infty} \frac{\ln^2 x}{x} = \lim_{x \rightarrow \infty} \frac{2 \ln x \cdot \frac{1}{x}}{1} = \lim_{x \rightarrow \infty} \frac{2}{x} = 0 \Rightarrow \lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{2\sqrt{x}}} = \lim_{x \rightarrow \infty} \frac{2}{\sqrt{x}} = 0$$

$$\therefore \lim_{x \rightarrow \infty} \frac{2 - \frac{\ln^2 x}{x}}{\frac{\ln x}{\sqrt{x}} + 1} = \frac{2}{1} = 2. \text{ Ans.}$$

Q.3 (C)

$$g(f^2(x)) = g(x^2 - 2x + 3)$$

$$\text{Now, } \frac{d}{dx} (g(x^2 - 2x + 3)) = g'(x^2 - 2x + 3) (2x - 2)$$

$$\therefore \left. \frac{d}{dx} (g(f^2(x))) \right|_{x=3} = g'(6) \cdot 4 = 4g'(6)$$

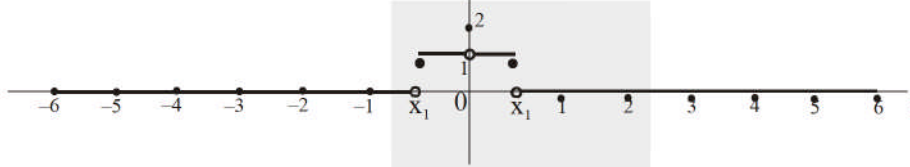
$$\text{Now, } g'(6) = \frac{1}{f'(1 + \sqrt{34})}$$

$$f'(x) = \frac{2(x-1)}{2\sqrt{(x-1)^2 + 2}} \Rightarrow f'(1 + \sqrt{34}) = \frac{\sqrt{34}}{6}$$

$$\therefore g'(6) = \frac{6}{\sqrt{34}} \Rightarrow 4g'(6) = \frac{24}{\sqrt{34}} \quad \text{Ans.}$$

Q.4 (C)

function is discontinuous at $x = 0, \pm x_1$ where $x_1 \in (0, 1)$



Q.5 (A)

$$t = x - \frac{1}{x}$$

$$t^3 = x^3 - \frac{1}{x^3} - 3t \Rightarrow f(t) = t^3 + 3t \Rightarrow f'(t) = 3t^2 + 3.$$

Q.6 (B)

$$f(x) = x - \sin x$$

$$f'(x) = 1 - \cos x$$

$$g(f(x)) = x \Rightarrow g'(f(x)) f'(x) = 1$$

$$\Rightarrow g'(f(x)) = \frac{1}{f'(x)}$$

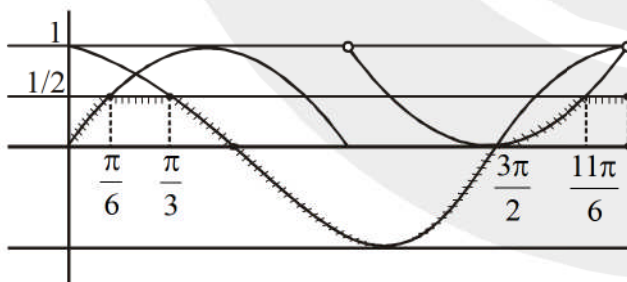
Clearly $g'(y)$ will not exist where $f'(x) = 0$

$$\Rightarrow 1 - \cos x = 0 \Rightarrow \cos x = 1 \Rightarrow x = 2n\pi, n \in \mathbb{I}$$

$$\text{at } x = 2n\pi, y = f(2n\pi) = 2n\pi$$

$\therefore$ Number of points of non-derivability in $(-10\pi, 10\pi)$ is 9. **Ans.**

Q.7 (D)



From the graph, $f(x)$ is non-differentiable at $x = \frac{\pi}{6}, \frac{\pi}{3}, \frac{3\pi}{2}, \frac{11\pi}{6}$

$$\therefore \sum x_i = \frac{\pi}{6} + \frac{\pi}{3} + \frac{3\pi}{2} + \frac{11\pi}{6} = \frac{23\pi}{6}$$

$\therefore \lambda = 23$. **Ans.**

Q.8 (C)

Since, $f(x)$ is continuous at $x=0$.

Hence, $K = \lim_{x \rightarrow 0^+} f(x)$; Put $x = t^2$

$$K = \lim_{t \rightarrow 0^+} \frac{e^{t^4} - \frac{2}{\pi} \sin^{-1} \sqrt{1-t^2}}{\frac{\ln(1+t)}{t} \cdot t}$$

$$K = \lim_{t \rightarrow 0^+} \frac{e^{t^4} - \frac{2}{\pi} \sin^{-1} \sqrt{1-t^2}}{t} \quad \left(\frac{0}{0} \right)$$

Using L-Hospital Rule

$$K = \frac{2}{\pi} \text{ Ans.}$$

Q.9 (B)

Q.10 (CD)

$$f(x) = x^2 + bx + c$$

$$\text{L.H.L., } \lim_{x \rightarrow 0^-} \frac{1}{\sqrt{2}} \left[\frac{\sin 2x}{x} \right] = \frac{1}{\sqrt{2}}$$

$$\text{R.H.L., } \lim_{x \rightarrow 0^+} \frac{e^{\sqrt{x-\sin x}} - f(6)x^{\frac{3}{2}} - e^{f(2)}}{(\tan x - x)^2} = \frac{1}{\sqrt{2}}$$

For limit to be exist $f(2) = 0$

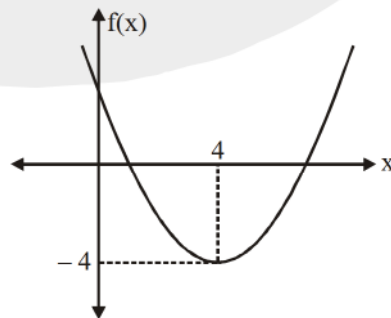
$$\lim_{x \rightarrow 0^+} \left(\frac{e^{\sqrt{x-\sin x}} - 1 - f(6) \cdot x^{\frac{3}{2}}}{\left(\frac{\tan x - x}{x^3} \right)^2 \cdot x^{\frac{3}{2}}} \right) = \frac{1}{\sqrt{2}}$$

$$\sqrt{3} \lim_{x \rightarrow 0^+} \left(\frac{e^{\sqrt{x-\sin x}} - 1}{x^{\frac{3}{2}}} - f(6) \right) = \frac{1}{\sqrt{2}}$$

$$\sqrt{3} \lim_{x \rightarrow 0^+} \left(\frac{\sqrt{x-\sin x}}{\sqrt{x^3}} - f(6) \right) = \frac{1}{\sqrt{2}}$$

$$\sqrt{3} \left(\frac{1}{\sqrt{6}} - f(6) \right) = \frac{1}{\sqrt{2}} \Rightarrow f(6) = 0$$

$$\therefore f(x) = x^2 - 8x + 12 = (x-4)^2 - 4$$



- (i) least value of $f(x)$ is -4
- (ii) $h(x) = |f(|x| + a) + b| = |(|x| + a - 4)^2 - 4 + b|$
 For $h(x)$ to be derivable for all $x \in \mathbb{R}$, $a = 4$ and $b \geq 4$.

Q.11 (AC)

Q.12 (ABD)

$$\lim_{x \rightarrow 0} \frac{f(2+h) - 2}{h} = 0 \text{ for limit to exist } f(2) = 2$$

$$\lim_{x \rightarrow 0} f'(2+h) = 0$$

$$\therefore f'(2) = 0 \quad \dots\dots(1)$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x) + f(h) - \frac{1}{x-1} - \frac{1}{h-1} + \frac{1}{x+h-1} - f(x)}{h}$$

Using L'Hospital Rule

$$f'(x) = \lim_{h \rightarrow 0} f'(h) + \frac{1}{(h-1)^2} - \frac{1}{(x+h-1)^2}$$

$$\therefore f'(x) = f'(0) + 1 - \frac{1}{(x-1)^2} \quad \dots\dots(2)$$

Put $x = 2$

$$\therefore f'(2) = f'(0) + 1 - 1$$

$$\Rightarrow f'(2) = f'(0) = 0 \text{ using (1)}$$

$$\Rightarrow f'(x) = 1 - \frac{1}{(x-1)^2}$$

$$\therefore f(x) = x + \frac{1}{x-1} + C$$

$$\text{as } f(2) = 2$$

$$\Rightarrow f(2) = 2 + 1 + C = 2$$

$$\therefore C = -1$$

$$\therefore f(x) = x - 1 + \frac{1}{x-1}$$

(i) $f'(x) = 1 - \frac{1}{(x-1)^2}$

$$f''(x) = \frac{2}{(x-1)^3}$$

- (A) $f''(2) = 2$; (B) Least value of $f(x)$ is 2

$$(C) \quad f'(2) + f(2) = 2 \quad (D) \quad \left. \frac{d}{dx}(f(x)) \right|_{x=f(2)=2} = f'(2) = 0$$

$\therefore$ Options (A) and (C) are correct.

$$(ii) \quad f(x) = 2 - \sin^2 \theta$$

$$\underbrace{x-1+\frac{1}{x-1}}_{\geq 2} = \underbrace{2-\sin^2 \theta}_{\leq 2}$$

The above equation will satisfy when $x = 2$ and $\sin \theta = 0$ i.e. $\theta = 0, \pi, 2\pi$

$\therefore [x + \theta] = 2, 5, 8.$

$$\text{Aliter: } f(x+y) = f(x) + f(y) - \frac{1}{x-1} - \frac{1}{y-1} + \frac{1}{x+y-1} + 1$$

Partially differentiating w.r.t. x

$$f'(x+y) = f'(x) + \frac{1}{(x-1)^2} - \frac{1}{(x+y-1)^2}$$

At $x = 2,$

$$f'(y+2) = f'(2) + 1 - \frac{1}{(y+1)^2}$$

Let $y+2 = t,$

$$f'(t) = 0 + 1 - \frac{1}{(t-1)^2} \quad (\because f'(2) = 0)$$

$$f(t) = t + \frac{1}{t-1} + c$$

At $t = 2,$

$$2 = 2 + 1 + c \Rightarrow c = -1 \quad (\because f(2) = 2)$$

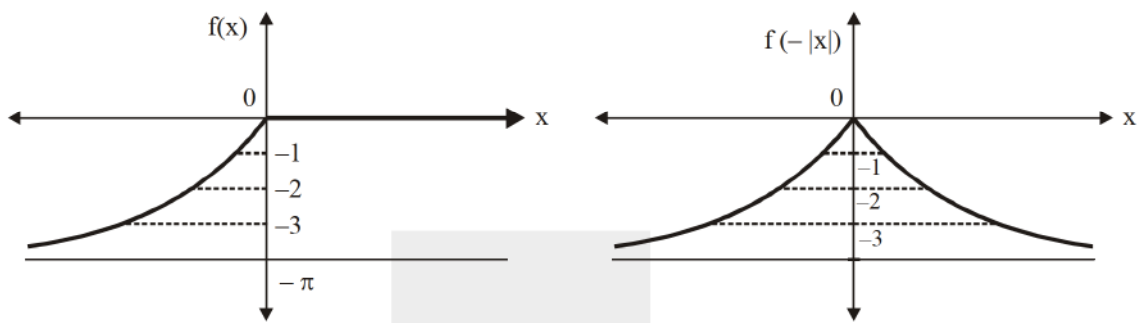
$$\therefore f(x) = x - 1 + \frac{1}{x-1}$$

Q.13 (AC)D)

$$f(x) = \sin^{-1} \left(\frac{2e^x}{1+e^{2x}} \right) - \pi + \cos^{-1} \left(\frac{1-e^{2x}}{1+e^{2x}} \right)$$

$$\text{Putting, } e^x = \tan \theta \Rightarrow \theta \in \left(0, \frac{\pi}{2} \right)$$

$$f(x) = \sin^{-1}(\sin 2\theta) - \pi + \cos^{-1}(\cos 2\theta) = \begin{cases} 4\theta - \pi; & \theta \in \left(0, \frac{\pi}{4} \right) \Rightarrow x < 0 \\ 0; & \theta \in \left[\frac{\pi}{4}, \frac{\pi}{2} \right) \Rightarrow x \geq 0 \end{cases}$$



Q.14 (AC)

Let $x^3 = t$

$$y^3 - ay^2t + byt^2 - ct^3 = 0 \begin{cases} y = m_1 t \\ y = m_2 t \\ y = m_3 t \end{cases}$$

Now, $y = mx^3$

$$\Rightarrow y' = 3mx^2 \Rightarrow y'' = 6mx \Rightarrow y''' = 6m$$

$$6m_1 = -6, 6m_2 = 12, 6m_3 = -18$$

$$m_1 = -1, m_2 = 2, m_3 = -3$$

$$\text{Let } \frac{y}{x^3} = m$$

$$\therefore \left(\frac{y}{x^3}\right)^3 + a\left(\frac{y}{x^3}\right)^2 + b\left(\frac{y}{x^3}\right) - c = 0$$

$$\therefore m^3 + am^2 + bm - c = 0 \begin{cases} m_1 \\ m_2 \\ m_3 \end{cases}$$

$$-a = -1 + 2 - 3 \Rightarrow a = 2$$

$$b = -2 - 6 + 3 = 5$$

$$\& c = -1 \times 2 \times -3 = 6.$$

Q.15 (AC)

$$\text{For } a = 0, l = \lim_{x \rightarrow 0} \frac{-x}{(\sin x)^b} = \begin{cases} b = 0 \rightarrow 0 \\ b = 1 \rightarrow -1 \end{cases}$$

$$\text{For } a = 1, \lim_{x \rightarrow 0} \frac{e^x - x - 1}{(\sin x)^b} = \lim_{x \rightarrow 0} \frac{\frac{x^2}{2!} + \frac{x^3}{3!} + \dots}{(\sin x)^b} = \begin{cases} b = 0 \rightarrow 0 \\ b = 1 \rightarrow 0 \\ b = 2 \rightarrow 1/2 \end{cases}$$

$$\text{For } a = 2, \lim_{x \rightarrow 0} \frac{1 + 2x + \frac{4x^2}{2!} + \dots - x - 1}{(\sin x)^b} = \begin{cases} b = 0 \rightarrow 0 \\ b = 1 \rightarrow 1 \end{cases}$$

$$\text{For } l_1 = 0, n_1 = 4$$

$$l_2 = 1, n_2 = 1$$

$$l_3 = -1, n_3 = 1$$



$$l_4 = \frac{1}{2}, n_4 = 1$$

Hence, $\sum_{i=1}^4 n_i = 7, \sum_{i=1}^4 l_i = \frac{1}{2}$ **Ans.**

Q.16 (AB)

$$2 \sin^{-1} x_0 + \tan^{-1} x_0 = \frac{5\pi}{\sqrt{-x_0^2 + 2x_0 + 15}}$$

$$\underbrace{2 \sin^{-1} x_0 + \tan^{-1} x_0}_{\leq \frac{5\pi}{4}} = \underbrace{\frac{5\pi}{\sqrt{16 - (x_0 - 1)^2}}}_{\geq \frac{5\pi}{4}}$$

$\therefore x_0 = 1$ is only solution

for $x = 1, y = \frac{2}{y+1}$

$$y^2 + y - 2 = 0 \Rightarrow (y+2)(y-1) = 0 \Rightarrow y = 1, -2$$

$$y^2 + xy = 2x \Rightarrow 2yy' + xy' + y = 2$$

$$y'(2y+x) = 2-y$$

$$y' = \frac{2-y}{2y+x}$$

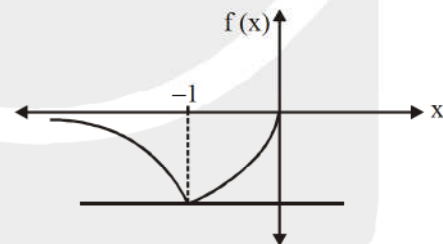
$$y'|_{(1,1)} = \frac{2-1}{3} = \frac{1}{3}$$

$$y'|_{(1,-2)} = \frac{4}{-3} \text{ Ans.}$$

Q.17 9

$f(x)$ is non-derivable at $x = -1, 0$

$[3x]$ is discontinuous at $x = \underbrace{\frac{1}{3}, \frac{2}{3}, \frac{3}{3}, \frac{4}{3}, \frac{5}{3}, \frac{6}{3}}_{5 \text{ points}}$



But $f(x)$ is non-derivable at 7 points

$\therefore f(x)$ should be continuous and derivable at $x = 2$

k must be 2

$$\therefore \lim_{x \rightarrow 2^-} f(x) = 5 = f(2)$$

$$(2-\alpha)^2 + \beta = 5$$

$$\text{and } f'(2) = 0 \Rightarrow 2(2-\alpha) = 0 \Rightarrow \alpha = 2$$

$$\beta = 5$$

$$k + \alpha + \beta = 9 \text{ Ans.}$$

Q.18 6

$$\lim_{x \rightarrow 0} \frac{(x - \sin x) \left(2x - \frac{\sin 2x}{2} \right) \left(3x - \frac{\sin 3x}{3} \right) \dots \left(nx - \frac{\sin nx}{n} \right)}{x^m} = 20$$

$$\lim_{x \rightarrow 0} \frac{\left(\frac{x - \sin x}{x^3} \right) \left(2 - \frac{\sin 2x}{2x} \right) \left(3 - \frac{\sin 3x}{3x} \right) \dots \left(n - \frac{\sin nx}{nx} \right)}{x^{m-(n+2)}} = 20$$

For limit to be exist

m must be equal to n + 2

$$\frac{1}{6} (2-1)(3-1) \dots (n-1) = 20$$

$$(n-1)! = 120 \Rightarrow n-1 = 5 \Rightarrow n = 6 \text{ Ans.}$$

Q.19 8

$$\log_2(f(3x)) = x + \log_2(3f(x))$$

$$\frac{f(3x)}{3f(x)} = 2^x$$

$$\frac{f(x)}{3 \cdot f\left(\frac{x}{3}\right)} \cdot \frac{f\left(\frac{x}{3}\right)}{3 \cdot f\left(\frac{x}{3^2}\right)} \cdot \frac{f\left(\frac{x}{3^2}\right)}{3 \cdot f\left(\frac{x}{3^3}\right)} \dots \frac{f\left(\frac{x}{3^{n-1}}\right)}{3 \cdot f\left(\frac{x}{3^n}\right)} = 2^{\frac{x}{3} + \frac{x}{3^2} + \dots + \frac{x}{3^n}}$$

$$\frac{f(x)}{3^n \cdot f\left(\frac{x}{3^n}\right)} = 2^{\left(\frac{x}{3} + \frac{x}{3^2} + \dots + \frac{x}{3^n}\right)}$$

$$\lim_{n \rightarrow \infty} \left(\frac{f(x) \cdot \frac{x}{3^n}}{f\left(\frac{x}{3^n}\right) \cdot x} \right) = \lim_{n \rightarrow \infty} 2^{\left(\frac{x}{3} + \frac{x}{3^2} + \dots + \frac{x}{3^n}\right)}$$

$$\frac{f(x)}{x \cdot f'(0)} = 2^{\left(\frac{x}{3} + \frac{x}{3^2} + \dots + \frac{x}{3^n}\right)}$$

$$f(x) = x \cdot 2^{\left(\frac{x}{2}\right)}$$

$$\therefore f(3) = 3 \cdot 2^{\left(\frac{3}{2}\right)} = 6\sqrt{2} \approx 8.4$$

$$\Rightarrow [f(3)] = 8 \text{ Ans.}$$



Q.20 0035

Putting $x = y = 0$, $f(0) = 0$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x) - f(x-h) + 3f(h) + 2xh}{2h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x) - f(x-h)}{2h} + \frac{3}{2} \lim_{h \rightarrow 0} \frac{f(h)}{h} + x$$

$$= \frac{1}{2} f'(x) + x \Rightarrow f'(x) = 2x \Rightarrow f(x) = x^2.$$

$$\therefore f(5) + f'(5) = 25 + 10 = 35. \text{ Ans.}$$

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